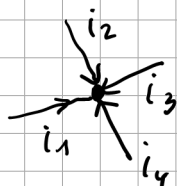


4 Kirchhoff'sche Regeln für Zeiger

Kirchhoff:

Knotenregel:



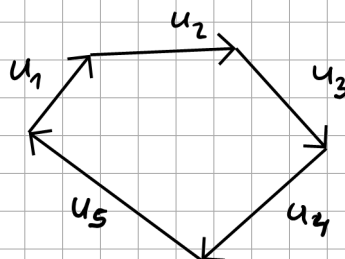
$$\sum_i i_i = 0$$

"KCL"

$$i_i(t) \rightarrow \underline{i}_i$$

$$\sum_i \underline{i}_i = 0$$

Maschenregel:

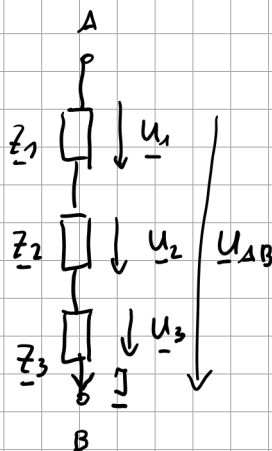


$$\sum_i u_i = 0$$

$$u_i(t) \rightarrow \underline{u}_i$$

$$\sum_i \underline{u}_i = 0$$

4.3 Reihenschaltung von Impedanzen



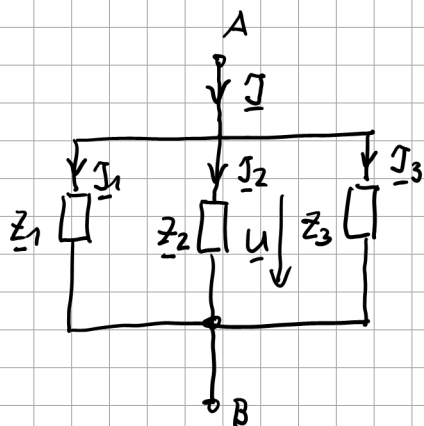
$$\begin{aligned} \underline{z}_{AB} &= \frac{\underline{u}_{AB}}{\underline{i}} = \frac{\underline{u}_1 + \underline{u}_2 + \underline{u}_3}{\underline{i}} \\ &= \frac{\underline{u}_1}{\underline{i}} + \frac{\underline{u}_2}{\underline{i}} + \frac{\underline{u}_3}{\underline{i}} \end{aligned}$$

$$\underline{z}_{AB} = \underline{z}_1 + \underline{z}_2 + \underline{z}_3$$

4.4 Parallelschaltung von Impedanzen (Admittanzen)

$\underline{Z}$ = Impedanz (komplexe Widerstand), reell: R

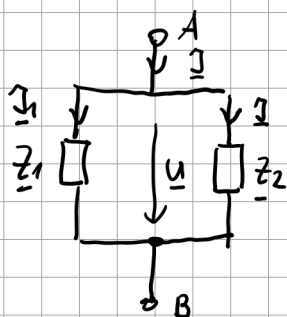
$\frac{1}{\underline{Z}} = \underline{Y}$ = Admittanz (komplexe Leitwert), reell: G



$$\begin{aligned}\underline{Y}_{AB} &= \frac{\underline{I}}{\underline{U}} = \frac{\underline{I}_1 + \underline{I}_2 + \underline{I}_3}{\underline{U}} \\ &= \underbrace{\frac{\underline{I}_1}{\underline{U}}}_{\underline{Y}_1} + \underbrace{\frac{\underline{I}_2}{\underline{U}}}_{\underline{Y}_2} + \underbrace{\frac{\underline{I}_3}{\underline{U}}}_{\underline{Y}_3}\end{aligned}$$

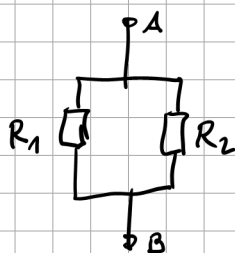
$$\underline{Y}_{AB} = \underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3$$

Falls 2 Impedanzen parallel:

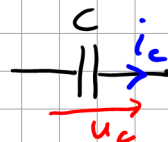
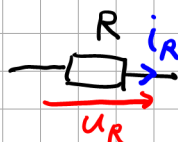


$$\begin{aligned}\underline{Z}_{AB} &= \frac{\underline{U}}{\underline{I}} = \frac{\underline{U}}{\underline{I}_1 + \underline{I}_2} = \frac{1}{\frac{\underline{I}_1}{\underline{U}} + \frac{\underline{I}_2}{\underline{U}}} = \frac{1}{\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}} \\ &= \frac{1}{\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}} = \frac{1}{\frac{\underline{Z}_2 + \underline{Z}_1}{\underline{Z}_1 \cdot \underline{Z}_2}}\end{aligned}$$

$$\underline{Z}_{AB} = \frac{\underline{Z}_1 \cdot \underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2}$$

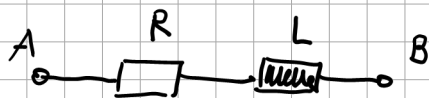


$$\Rightarrow R_{ges} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$



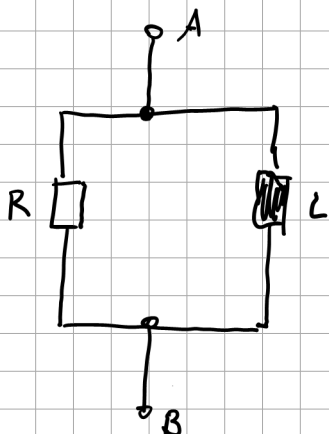
Einheiten	$[R] = \Omega = \frac{V}{A}$	$[C] = F = \frac{As}{V}$	$[L] = H = \frac{Vs}{A}$
Gespeicherte Energie	/	$W_C = \frac{1}{2} C u^2$	$W_L = \frac{1}{2} L i^2$
Zeitbereich	$u_R(t) = R \cdot i_R(t)$	$i_C(t) = C \cdot \frac{du(t)}{dt}$ $u_C(t) = \frac{1}{C} \int i_C(t) dt + u_0$	$u_L(t) = L \frac{di_L(t)}{dt}$ $i_L(t) = \frac{1}{L} \int u_L(t) dt + i_0$
Frequenzbereich	$\underline{U}_R = R \cdot \underline{I}_R$	$\underline{U}_C = \frac{1}{j\omega C} \underline{I}_C$	$\underline{U}_L = j\omega L \cdot \underline{I}_L$
Impedanz	$\underline{Z}_R = R$	$\underline{Z}_C = \frac{1}{j\omega C}$	$\underline{Z}_L = j\omega L$
Admittanz	$\underline{Y}_R = \frac{1}{R}$	$\underline{Y}_C = j\omega C$	$\underline{Y}_L = \frac{1}{j\omega L}$
Reaktanz $\hat{=}$ Impedanz	$X_R = R$	$X_C = \frac{1}{\omega C}$	$X_L = \omega L$
Zeiger			

Beispiele:



$$\underline{Z}_{AB} = \underline{Z}_R + \underline{Z}_L = R + \underbrace{j\omega L}_{X_L} = R + j\underbrace{X_L}_{\text{Reaktanz}}$$

Resistanz Reaktanz



$$\underline{Z}_{AB} = ?$$

Parallelschaltung:

$$\underline{Y}_{AB} = \underline{Y}_R + \underline{Y}_L$$

$$= \frac{1}{R} + \frac{1}{j\omega L}$$

$$\underline{Y}_{AB} = \frac{j\omega L + R}{j\omega L \cdot R}$$

$$\underline{Z}_{AB} = \frac{1}{\underline{Y}_{AB}} = \frac{j\omega L R}{R + j\omega L}$$

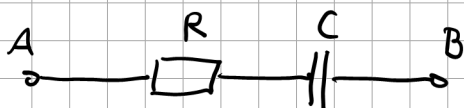
$$\underline{Z}_{AB} = \frac{\underline{Z}_R \cdot \underline{Z}_L}{\underline{Z}_R + \underline{Z}_L} = \frac{R \cdot j\omega L}{R + j\omega L} \cdot \frac{(R - j\omega L)}{(R - j\omega L)}$$

$$(a + jb)(a - jb) = a^2 - j^2 b^2 = a^2 + b^2$$

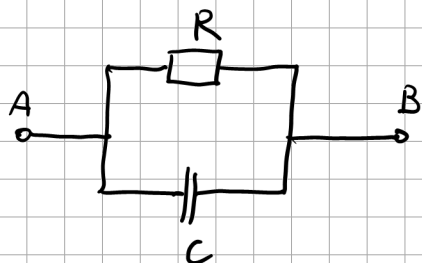
$$= \frac{j\omega L R^2 + \omega^2 L^2 R}{R^2 + (\omega L)^2}$$

$$= \frac{\omega^2 L^2 R}{R^2 + (\omega L)^2} + j \frac{\omega L R^2}{R^2 + (\omega L)^2}$$

Übungsaufgaben



$$\underline{Z}_{AB} = \dots$$



$$\underline{Z}_{AB} = \dots$$

$$R = 50 \Omega \quad C = 1 \text{ mF} \quad f = 50 \text{ Hz}$$

$$\underline{Z}_{AB} = \underline{Z}_R + \underline{Z}_L$$

$$= R + \frac{1}{j\omega C}$$

$$\boxed{\frac{1}{j} = -j}$$

$$= R - j \frac{1}{\omega C}$$

$$\omega = 2\pi f$$

$$= 50 \Omega - j \frac{1}{2\pi \cdot 50 \cdot \frac{1}{5} \cdot 1 \text{ mF}}$$

$$[F] = \frac{As}{V}$$

$$= 50 \Omega - j \frac{1}{2\pi \cdot 50 \cdot \frac{1}{5} \cdot 10^{-3} \frac{As}{V}}$$

$$= \underbrace{50 \Omega}_R - j \underbrace{3,18 \Omega}_{X_C}$$

$$\underline{Z}_{AB} = \frac{\underline{Z}_R \cdot \underline{Z}_C}{\underline{Z}_R + \underline{Z}_C}$$

(Parallelschaltung)

$$= \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{\frac{R}{j\omega C}}{\frac{j\omega C R + 1}{j\omega C}}$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \cdot \frac{d}{c}$$

$$= \frac{R}{1+j\omega RC} \cdot \frac{1-j\omega RC}{1-j\omega RC}$$

$$= \frac{R - j\omega R^2 C}{1^2 + (\omega RC)^2}$$

$$= \underbrace{\frac{R}{1 + (\omega RC)^2}}_{\text{Realteil}} - j \underbrace{\frac{\omega R^2 C}{1 + (\omega RC)^2}}_{\text{Imaginärteil}}$$

$$= \frac{50 \Omega}{1 + (2\pi \cdot 50 \frac{1}{s} \cdot 50 \frac{V}{A} \cdot 10^{-3} \frac{As}{V})^2} - j \cdot \frac{2\pi \cdot 50 \frac{1}{s} \cdot (50 \Omega)^2 \cdot 10^{-3} \frac{As}{V}}{1 + (2\pi \cdot 50 \frac{1}{s} \cdot 50 \frac{V}{A} \cdot 10^{-3} \frac{As}{V})^2}$$

$$= \frac{50 \Omega}{247,7} - j \frac{15,7 \Omega}{247,7}$$

$$= 0,202 \Omega - j 0,063 \Omega$$